Coolfont Mountainside Association

Cross-Connection & Backflow Prevention Policy Overview

Why This Matters

State regulations require CMA to implement and comply with a Cross-Connection and Backflow Policy to reduce the risk of contamination of the water system. Homeowner participation is essential to help achieve compliance.

Our Shared Responsibility

Protecting our water ensures safe drinking water, regulatory compliance, and long-term sustainability. We all play a role in keeping our system safe. This policy affects all homeowners.

So What Is a Cross-Connection?

A Cross Connection is a link between a public water source and a source of potentially contaminated water.

Examples:

- A direct connection – one pipe to another pipe.
- A submerged inlet connection where incoming water enters a container below the water level.
- A hose submerged in a hot tub.

What Is Backflow?

Backflow occurs when water flows in the wrong direction, potentially bringing contaminants into the water supply. This can happen when pressure inside your home exceeds system pressure, such as with pumps or pressure systems.

Examples:

- A rain barrel connected to the CMA water system
- A hose submerged in a hot tub, pool, or bucket

Do's

- **Install anti-siphon valves**
- **Use hose bib vacuum breakers**
- **Maintain all backflow prevention devices**
- **Uninstall the hose bib in the winter when temperatures drop below freezing and reinstall it in the spring when temperatures rise**

These devices are inexpensive and can be found online or at local hardware stores.

Don'ts

- **Leave a hose submerged in a liquid, including while filling a hot tub**
- **Use a garden sprayer (with chemicals) without proper backflow protection. Hose bib vacuum breakers will provide protection**
- **Leave your hose bib backflow preventers on if they aren't freeze-proof. Instead, remember to remove and replace it seasonally!**

What Happens Next

The Board approved the policy on April 20, 2026. A homeowner survey will be distributed and must be completed within 90 days. Moving forward, surveys will be sent out to be updated with septic maintenance notifications.

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Cross-Connection & Backflow Prevention Policy Helpful Definitions

What Is a Cross-Connection?

A cross-connection is any actual or potential connection between clean drinking water (potable water) and a non-potable water source. If backflow occurs, contaminants can enter the community water supply and create serious public health risks.

Examples of possible cross-connections include:

- Irrigation systems
- Swimming pools and hot tubs
- Outdoor hose bibs
- Fire sprinkler systems
- Chemical or industrial systems

Because plumbing systems are continually being installed, modified, and extended, ongoing monitoring and prevention are required.

Preventing Cross-Contamination

The most effective way to prevent contamination is by installing appropriate backflow prevention devices. These devices stop contaminated water from flowing backward into the potable water supply.
Below are common types of backflow prevention devices and where they are typically used.

Vacuum Breaker (Hose Bib)

A vacuum breaker is a small, inexpensive device that screws onto an outdoor hose bib to prevent contaminated water from siphoning back into your drinking water supply. These are generally easy for homeowners to install themselves.

Example: Vacuum Breaker Installation Video

Pressure Vacuum Breaker (PVB)

A Pressure Vacuum Breaker prevents contaminated water from hot tubs, irrigation systems, or outdoor plumbing from flowing back into the potable water supply. It uses a spring-loaded check valve and air inlet to prevent back-siphonage. Many DIY homeowners can install these themselves.

Example: Pressure Vacuum Breaker Installation Video

Double Check Valve Backflow Preventer

A double check valve assembly uses two internal check valves to prevent backflow. These systems are commonly used where testing and inspection are required to verify proper operation. Installation is more complex and may require professional assistance.

Example: Double Check Valve Installation Video

Reduced Pressure Zone (RPZ) Device

An RPZ device is a high-hazard backflow preventer designed to protect drinking water from hazardous contaminants. It uses two check valves and a relief valve to prevent contamination if either valve fails. RPZ devices are commonly required for commercial, irrigation, or other high-risk applications.

Important: West Virginia regulations (64CSR15) require RPZ devices to be installed by a qualified, licensed professional and tested by a West Virginia Certified Backflow Prevention Device Tester.

Example: RPZ Device Installation Video

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Cross-Connection & Backflow Prevention Policy Frequently Asked Questions

Do I need to be concerned about my washing machine for backflow or cross-connection?
No. Washing machines do not typically have pressure connections that pose a backflow risk under standard policies.

Do I need to be concerned about my rain barrel?
It depends on whether the barrel is connected to the CMA water system through your home's plumbing.

- Most rain barrels are not connected to the water system and do not require a backflow device.
- If your rain barrel is connected, the need for a device depends on its location and how it's connected.

If you're unsure, have it reviewed by a CMA Water & Sewer representative or a qualified plumber.

I have a water softener—what do I need to do?
Most water softeners do not require a backflow prevention device, as they are typically designed with built-in protections. If your system was professionally installed and has not been modified, it is likely fine.

If you're unsure about the installation, notice any modifications, or see anything unusual, have it inspected by a West Virginia–licensed plumber familiar with backflow requirements.

Will the Water & Sewer Committee be installing the devices?
No. Installation is the homeowner's responsibility. In many cases, a simple hose bib vacuum breaker (air gap device) is sufficient. These cost about $10 or less and can be installed by the homeowner.

If additional protection is needed or you're unsure, CMA recommends working with a West Virginia–licensed plumber experienced in backflow prevention.

How could a hose in my hot tub contaminate the water system?
Backflow occurs when water pressure reverses. If the CMA system pressure drops below the pressure in a connected system (even slightly), water can flow backward—potentially pulling contaminated water (e.g., from a hot tub, chemicals, or pesticides) into your home or the shared system.

While rare, there are documented cases of illness and even death caused by backflow contamination. That's why prevention is required.

Do I need to add anything to my toilets?
Most modern toilets (manufactured after 1973) either:

- Do not require additional protection, or
- Already have built-in backflow prevention

Do I need a backflow preventer with my septic tank?
Only if you have a septic system pumping effluent uphill to the CMA effluent collection system

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